

1.1 Introduction	1
1.2 Chosen Idea.....	1
1.3 Other ideas	1
1.3.1 Buckshot Roulette	1
1.3.2 Super Meat Boy.....	2
1.3.3 Osu:	2
1.3.4 Clash Royale	2
2.1. Game Features and Gameplay.....	3
2.1.1 Story.....	3
2.1.2 Levels.....	3
2.1.3 Items	4
2.1.4 Enemies	5
2.2. Game Design Principles.....	5
2.2.1. Pursuing and Achieving Goals	5
2.2.2. Game Interactivity.....	6
2.2.3. Feedback	6
2.2.4. Gameplay Variety.....	6
2.2.5. Gameplay Consistency and Fairness	7
2.2.6. Avoid Repetition.....	7
2.3. Game Rules	7
2.3.1. Constitutive.....	7
2.3.2. Operational	8
2.3.3. Implicit	8
3. Software Engineering.....	8
3.1. Software Requirements.....	8
Functional Requirements	8
Non-Functional Requirements:	10
3.2. Acceptance Tests	11
4. Implementation Plans	14
4.1. Task List	14
4.2. Activity Network with Critical Path	17
4.3. Gantt Chart	17

1.1 Introduction

After the first workshop, we got to know each other through the first task we completed together. Then, we started thinking of ideas for a game. Everyone had to come up with at least one game that we could create for our project. Then, we met up and voted on the idea we all liked best and thought was possible to create. Then, everyone had to create a story for the game, and we voted on the one we liked the most.

1.2 Chosen Idea

We decided on a roguelike game. You walk around levels and shoot waves of enemies. After defeating all the enemies on a floor, you must defeat the boss for that level. Each level is harder, with enemies that are stronger and deal more damage. However, the player can collect items and upgrades to become stronger and match the enemies' strength. The player fights with alien weapons scattered around the ship while trying to save the other humans.

Key features:

- Space/alien-themed roguelike game
- Monsters that follow and shoot the player
- Fight waves of enemies
- Hard boss fights

1.3 Other ideas

1.3.1 *Buckshot Roulette*

A strategy simulator game in which you play against an opponent. You can obtain items that give you an advantage, but they can also backfire on your strategy. It is an unpredictable game in which you must fight the odds to emerge victorious.

Key features:

- Fight the odds
- Calculate your chances of victory
- Important decision-making
- High-pressure game with suspenseful moments

We decided against this idea because it limits repeatable gameplay. Updating the game might make it feel too complicated or overcrowded. Also, our group could not think of any unique features to add to the game.

1.3.2 *Super Meat Boy*

Super Meat Boy is a 2D platformer game in which you jump over obstacles to reach the finish line and advance to the next level. Players can earn stat bonuses for the current level or overall upgrades to make it easier to reach the next level.

Key features:

- 2D side-view gameplay
- Skill-based gameplay
- Every level is unique.
- You can choose different routes.

This game seemed like a good choice because it was not repetitive. However, the bigger problem was the creativity required to design all the levels, as well as the time needed for game testing to ensure the game was playable and that players could complete the levels. This would have meant that we would have needed to spend more time doing other things besides coding.

1.3.3 *Osu:*

This is a rhythm-based game in which players click on blocks or circles on the screen to earn points. You get bonus points if you keep a streak of not missing targets and if you hit them precisely on the beat.

Key features:

- Rhythm-based game
- Unique levels for every song or beat
- Get a score after completing a level
- Timing-based gameplay

After a long discussion, we decided that this kind of game would require a lot of design and creative implementation. We would need to create or find a copyright-free song and design a level for it. To avoid making the game seem repetitive, we need to add many levels so that players can experience diversity. This forces us to have more creative sessions where we design levels and code features into the game.

1.3.4 *Clash Royale*

Clash Royale is a strategy MOBA-type game in which you place characters on a map. Your objective is to destroy the enemy base. The characters have different abilities and stats, so it is important to consider how and when you play them. You must also decide which characters to include in your team so they can work together on the map.

Key features:

- Strategy-based game
- Playing against AI

- Every round can be different.
- 1-on-1 format

We decided not to create this type of game because it would require extensive NPC and AI movement and decision-making. We were not sure how difficult this task would be.

2.1. Game Features and Gameplay

2.1.1 Story

The player has the chance to see what the astronaut X goes through and control him. Sent in a revolutionary mission to check if HD 137010 b is habitable, X was suddenly blinded by a flash of bright light, making him lose control of his spaceship, then everything went black. The next thing he knows is waking up in the docking bay of an unknown spaceship. The story focuses on X trying to find a way to escape and repair his spaceship to be able to contact the base and return home. The game becomes more intense as the player gets closer to the final room of the spacecraft, where they are faced with a shocking truth about the reason why they were captured, the aliens are conducting several experiments on humans in order to create mutants to form an army and capture the neighbour planets.

A cutscene will be played at the beginning of the first level (the player has the option to skip it), to present the events that happened right before the crash so that the player has a clear idea of what led to the situation they will find themselves in at the start of the game, and give a clear view of the game objective.

2.1.2 Levels

The game is divided into four levels, each of them representing a different room in the alien base. In each level, the player will have to fight against several numbers of aliens, with different visuals and abilities, that will become stronger as the game progresses. The player will have a set time to complete the level, that will show clearly so the player will be able to check at any point how much time they had left to complete the level. A level will end as soon as the player defeated all the enemies, they run out of hearts (health system) or if they run out of time. The player will finish the game after they successfully passed through all the rooms.

The first level will act as a tutorial; after the introductory cutscenes ends, the player will be directly in the first room. The controls will be displayed on the floor for a few seconds so that the player will know what each key does and how the mouse-based combat works. In this level, the difficulty will be low, meaning that the enemies will be weak and predictable. Thus, the player has the chance to get familiar with the aiming and movement controls before diving into more intense scenarios. Additionally, there will be some base items the player can collect, such as the weapon, spaceship parts and some stats-enhancing objects that they can trade in later levels. In order to unlock the door of the next level, the player has to defeat the secondary enemies and then the mini-boss (this applies to all levels), which will be slightly stronger than the previous enemies, but still weak enough for the player to easily defeat it.

With the second level, the difficulty increases, as the enemies will be stronger; and more features will be introduced. Starting with this level, at the beginning of each level, the player will have the chance to trade one of the previously collected items with another one to upgrade their weapon or special abilities. Also, the enemies will

come in waves, multiple of them attacking the player at the same time, which means that the player needs to pay more attention and put in more effort to complete the level before they run out of time while managing to not run out of health.

The third level will have a slightly different visual scheme compared to the others; there will be some hints regarding the final. This level focuses on enemy variety and tactics. The waves of enemies will be more frequent, and the enemies will be faster. To complete this level, the player will have to engage continuously in combat and try to be as efficient as possible. In this level, the special abilities will be extremely useful if they are used properly, the same goes for the weapon if it was upgraded in the previous levels during the trading.

The final level will be the hardest and the most diverse. All the previous enemy categories will be present. Rather than one alien type waves, the player will be ambushed by different types of aliens at the same time, so accurate shooting and efficiently dodging the aliens' attacks is crucial. At the end of the level, the player will have to use all of their knowledge of the game mechanics to defeat the final boss, which will have the highest difficulty among the previously encountered aliens.

2.1.3 Items

Items are designed with variety in mind. There will be standard items such as health kits to provide support during encounters with the enemies, balancing the difficulty and enabling the player to recover after small mistakes.

However, most items will be inspired by space and alien technology. These items include: laboratory equipment (e.g. test tubes with different substances/ syringes) that will temporarily increase the player's speed, alien plant or speedometer that will boost fire rate and electronic/ magnetic (e.g. radio) devices that might be able to interfere with the enemy's mechanics, making them an easier target.

The game will have a trading system at the beginning of each level, starting from the second one. The player will have the chance to exchange base items for more powerful ones, that can either upgrade their weapon or improve their special abilities.

The spaceship parts will not affect combat in any way, and the player will not be able to use them during the trading, but they will be important for progression. Collecting them will encourage the player to explore more rather than focusing solely on combat, while keeping the core of the game intact.

2.1.4 Enemies

The design of enemies concentrates on clear roles and understandable behaviour. At the beginning of the game, the enemies are weak, somewhat aggressive and easy to defeat, giving the player the chance to familiarise themselves with the aiming and movement systems in low difficulty situations, reducing frustration.

As the game progresses, enemies will be harder to defeat as their stat will increase. There will be slower, strongly built aliens that will force the player to be patient and focus more on movement and positioning, while faster enemies will increase the combat intensity, making the attention and reaction time crucial.

Later in the game, the enemies will combine traits from previous levels to increase difficulty, while keeping the core mechanics of the game. The mutants present in the final reflect the game's theme and act unpredictably to keep the player engaged.

To advance to the next level, the player will have to defeat a mini boss, whose stats will be higher than the ones of the secondary enemies present in that level. This type of bosses will be like the other enemies but will act more aggressively and constantly attack the player. However, they will still be weaker than the final boss.

The final boss design will take into considerations the strengths and weaknesses of the previous enemies to signal the player that they reached the culmination point of the story. It would use several attack patterns, the player being able to defeat it only if the player managed to "master" movement, aim and ability usage.

2.2. Game Design Principles

2.2.1. Pursuing and Achieving Goals

The game is designed so that the player always has clear goals to achieve. From the moment the game starts with the player waking up after the spaceship crash, the player is placed in a hostile environment where survival and progression are the main objectives. Each room is a challenge the player must complete to progress further into the ship and trying to reach the final room.

The goals of the game are structured at different levels:

- **Short term goals:**
Survive each individual room. This involves defeating the enemies in the room, avoiding damage, try not to lose a life early in the game and using items or abilities effectively in order to stay alive.
- **Medium term goals:**
Over time, the player aims to progress through a series of rooms. As progress is made, the enemies gradually become more challenging, and fights will require quicker reaction, better positioning and more careful use of the abilities.
- **Long term goals:**
The ultimate objective of the game is to reach the final room of the spaceship, defeat the final and stronger boss so that you stop the aliens from experimenting on humans by freeing the captured survivors.

The difficulty of the game is set so that the game is neither too frustrating nor easy for the player. As the player progresses, they will encounter stronger enemies but on the other hand, the player gets the availability of upgrades, items as well as special abilities which strengthens the chances to win.

2.2.2. Game Interactivity

The game has a strong emphasis on player interaction, meaning that the player is never a passive observer. Using the mouse to shoot allows for direct and continuous player involvement since it relies on precise aiming, timing and quick reactions.

Decisions such as how to position themselves in a room, when to use special abilities and which items to collect or use can determine whether the player survives or fails. The feature of abilities with cooldowns rather than using them repeatedly, adds even more interactivity. So how the game turns out is depended directly on the player's decisions.

In addition, the variety of enemy types found in each room encourage the player to adapt their playstyle instead of relying on the same approach every time. As a result, gameplay always remains engaging and responsive, with the player needing to react to different situations as they arise.

2.2.3. Feedback

The feedback of the game is continuous and clear in order to help the player understand their current state and progress. Visual feedback such as enemy animations, health bar, damage indicators and cooldown timers for abilities ensures that the player always knows the current situation.

There is also constant feedback on the player's progress relative to their progress. Beating a room of enemies, defeating a boss or reaching the final room makes the user feel that they accomplished something. In addition sound effects also support this, as enemy attacks and special abilities usually have clear audio cues that indicate what is happening.

In general, the game interface is kept simple, so the player is not expected to memorise information or search through complex menus. Most information needed is already visible allowing the player to focus on gameplay rather than on understanding the interface.

2.2.4. Gameplay Variety

- The player must use different combat strategies depending on the room layout and the types of alien enemies encountered aboard the spaceship.
- Early levels allow simple strategies to be effective, while later levels introduce stronger enemies and more complex room designs that require the player to adapt their approach and try new strategies.
- New abilities, items, and enemy behaviors are introduced gradually as the player progresses, increasing the range of possible decisions without overwhelming the player.

- Low-level mechanics, such as enemy behavior and basic combat actions, are handled automatically by the game, allowing the player to focus on movement, positioning, and upgrade choices.

2.2.5. *Gameplay Consistency and Fairness*

- There will be consistency in gameplay actions and outcomes, because:
 - The player will always fail by losing all health due to alien enemies or environmental hazards.
 - Enemies behave consistently based on their type, allowing players to learn patterns and plan their actions.
 - Rooms will only be cleared once all enemies have been defeated, and defeated enemies will not reappear.
- The game will be fair, as all enemies can be defeated using the player's available weapons and abilities.
- Difficulty will increase gradually as the player progresses through the spaceship, through stronger enemies and more complex room layouts rather than sudden or unfair difficulty spikes.
- Story elements, including the final experimentation room and narrative reveal, will align with the gameplay mechanics already established, maintaining immersion and avoiding contradictions.

2.2.6. *Avoid Repetition*

- The player will be able to skip cutscenes they have already seen, such as introductory or story-related sequences.
- The player will be able to save their progress, preventing the need to replay completed sections of the game.
- Repetition will be reduced through procedurally generated rooms, a variety of alien enemies, and multiple upgrades and abilities available to the player.
- The player will not be required to wait for actions to complete, as movement, combat, and abilities will respond immediately.

2.3. Game Rules

2.3.1. *Constitutive*

- Player's health is set to 100% (number of hearts)
- Enemies spawn when player enters the room (secondary enemies and a main enemy after the secondary ones die)
- Enemies are removed from the game when their health bar reaches zero.
- The player can move freely in the background.
- The player can attack the enemies using the primary, secondary, or special attack.
- The player gets damaged whenever the enemy attack hits.
- Each level is completed when all the enemies (secondary and main) are defeated.

- The door to the next room is locked until the current room/level is completed.
- The main enemy of each room will be more powerful and have more health than the secondary enemies.
- The player needs to pass the level before a countdown reaches zero (e.g 2 min countdown).

2.3.2. *Operational*

- The player's movement is controlled with the keyboard and the mouse.
- The power of the enemies and the level of difficulty increase as the player completes the rooms.
- The player can collect items dropped by the enemies.
- The player can trade the collected items with other weapons, skills, or skins for the next level/room.
- Each room will contain different enemies and structures, to make it harder for the player to win.
- The game can be paused at any time (except for cinematics).
- The health of the player and other stats are displayed on the screen during the match/level.
- Animation of movement, and damage for player and enemies reflects what is happening.

2.3.3. *Implicit*

- Correlation between more powerful attacks and more health for the enemies.
- Progressive difficulty for the different levels/rooms.
- Avoiding enemy attacks is necessary to survive the level/room.
- Structures in the space will block the movement of the player, making it harder for the player to evade enemy attacks.
- Visual and audio details will show the player change of state (e.g. the main boss is coming, the player has almost no health left, etc)
- The player learns attack patterns of the enemies with the experience.
- The player can collect items dropped by the enemies to exchange them for skills, skins, or weapons for the next rooms.
- The level gets restarted when the player dies before completing the room.
- Different structures do different things (e.g. blocking player movement, causing damage, slowing down the player, etc.)

3. Software Engineering

3.1. *Software Requirements*

Functional Requirements

F1: The game shall display important gameplay information through the GUI, including:

- F1.1 Player Health.
- F1.2 Remaining time to pass the level (countdown).
- F1.3 Current level/room number.
- F1.4 Player's currently active abilities, and cooldown times.
- F1.5 Player's current equipped weapon/skill.
- F1.6 Collected items.

F2: The game shall allow the user to control the character by input from the keyboard and mouse.

- F2.1 Player movement shall be controlled by keyboard input.
- F2.2 Player direction (aim) shall follow the mouse pointer.
- F2.3 Main player attack action shall be performed by mouse left click.
- F2.4 Player special attack shall be triggered by keys 'Q' and 'E'.

F3: The game shall include multiple enemy types that behave differently during gameplay.

- F3.1: Some enemies shall move faster than others.
- F3.2: Some enemies shall deal higher damage or have increased health compared to basic enemies.

F4: The game shall allow the player to progress through multiple rooms or levels.

- F4.1: Each room shall only be considered completed once all enemies are defeated.
- F4.2: The game shall increase difficulty as the player progresses through rooms.

F5: The game shall provide audio feedback for important gameplay events.

- F5.1: Sound shall play when the player uses a special ability or when the player attacks and defeats an enemy.
- F5.2: Sound shall play when the player takes damage or is defeated by an enemy.

F6: The game shall allow the player to pick up items during gameplay.

- F6.1: Collected items shall affect the player's gameplay, such as survivability or improving abilities.

F7: The game shall handle player death and level failure.

- F7.1: When the player's health reaches zero the current level shall end.
- F7.2: The game shall allow the player to be able to restart the level after death.

F8: The game should spawn enemies at random locations on the map but not directly on top of the player.

F9: The enemies shall walk towards the player, but it should not be impossible for the player to run from them.

F10: Enemies shall disappear after they die.

F11: After defeating all the waves of enemies, the final boss for the room should spawn.

F12: The game shall end after defeating the last boss in the last room.

F13: The game shall allow the player to trade the items collected in the previous levels.

F14: The game shall spawn unbreakable static objects on the map but not directly on top of the player.

- F14.1: The objects shall block player or enemy movement.
- F14.2: The objects shall remain in the same position for the entire duration of each level.
- F14.3: The objects shall block attacks from both the player and the enemies.
- F14.4: The objects shall not be destroyed by any form of attack.

F15: The game shall show a cutscene at the beginning of the first level.

- F15.1: The game shall allow the cutscene to be skipped.

F16: The player shall be able to pause the game at any time, except during a cutscene.

- F16.1: The pause menu shall show options for continuing or restarting the game.

Non-Functional Requirements:

NF1: The Videogame shall be runnable on macOS, Windows, and Linux operating systems.

NF2: The Videogame shall run at 60 frames per second consistently on the average hardware and system, and not to drop framerate below 30 frames per second.

NF3: The Videogame shall be responsive to player input (keyboard and mouse) within 100ms during gameplay.

NF4: The Videogame shall handle errors without breaking, and display feedback information (Logs) to the user whenever it is possible.

NF5: The Videogame structure should be aimed for future maintainability, scalability, and debugging.

NF6: The Videogame shall support full screen mode, while maintaining the aspect ratio.

NF7: The Videogame shall load each new level within 7 seconds on average hardware/system.

NF8: The Videogame shall allow the user to modify the sensibility of the mouse.

NF9: The Videogame shall have a log system at runtime for debugging and development purposes.

NF10: The Videogame shall start the menu within 5 seconds after file execution.

3.2. Acceptance Tests

ID	Description	Input(Method)	Expected Output
System			
NF1	The Videogame shall be runnable on macOS, Windows, and Linux operating systems.	Run the game on all three operating systems.	Game runs and is playable.
NF3	The Videogame shall be responsive to player input (keyboard and mouse) within 100ms during gameplay.	Start the level.	Player moves and shoots.
		Check if every player key does its function.	
NF4	The Videogame shall handle errors without breaking, and display feedback information (Logs) to the user whenever it is possible.	Player overwrites the code in some way.	The game does not stop or lose progress.
NF5	The Videogame structure should be aimed for future maintainability, scalability, and debugging.	Add more items and abilities.	The game is still playable.
NF7	The Videogame shall load each new level within 7 seconds on average hardware/system.	Start the first level.	Every level loads.
		After beating it go into the second level.	
NF8	The Videogame shall allow the user to modify the sensibility of the mouse.	Edit mouse sensibility.	Mouse moves slower or faster.
NF9	The Videogame shall have a log system at runtime for debugging and development purposes.	Run the game.	Check the status of the game.
NF10	The Videogame shall start the menu within 5 seconds after file execution.	Start the game.	Main screen is reached under 5 seconds.
Graphics - General			
NF6	The Videogame shall support full screen mode, while maintaining the aspect ratio.	Enter full screen.	Game does not crash or dimorphs objects.
NF2	The Videogame shall run at 60 frames per second consistently on the average hardware and system, and not to drop framerate below 30 frames per second.	Load maximum number of enemies and have maximum number of projectiles.	Game runs and is playable.
Graphics - GUI			
F1	The game shall display important gameplay information through the GUI, including:		
F1.1	Player Health.	Start a level.	Health Metric in the form of UI showing 100% of players health
F1.2	Remaining time to pass the level (countdown).	Start a level.	Countdown timer is visible to the player showing the remaining time to complete the level.
F1.3	Current level/room number.	Start a level.	UI shows the current level that the user is playing during the whole match.
		The level is completed.	UI is updated to the next level (number).

F1.4	Player's currently active abilities, and cooldown times.	Start a level.	Currently active equipment/abilities of the player are shown in UI during the match.
F1.5	Player's current equipped weapon/skill.	Player uses a special ability.	Show special-ability cooldown in the UI during the match
F1.6	Collected items.	Player picks up item.	Show item in the inventory of items during the match
Levels			
F4.1	Each room shall only be considered completed once all enemies are defeated.	Start the room/level with all the enemies active and defeat them.	The room/level is marked as completed, and the player progresses to the next level.
F7.1	When the player's health reaches zero, the current level shall end.	The player takes continuous damage until health reaches zero.	The level immediately ends, and a defeat screen is displayed.
F11	After defeating all the waves of enemies, the final boss for the room should spawn.	Defeat enemies for the level	Final boss of the level spawns
F12	The game shall end after defeating the last boss in the last room.	Defeat the last boss in the final room	The game ends
F13	The game shall allow the player to trade the items collected in the previous levels.	The player selects the item they want to trade and the item they want to get in return.	The new item is added to the list of collected items.
F15	The game shall show a cutscene at the beginning of the first level.	Start the game.	The cutscene will play.
F15.1	The game shall allow the cutscene to be skipped.	During the cutscene, the player selects 'skip'.	The cutscene will stop and the player will be able to play.
Difficulty			
F3	The game shall include multiple enemy types that behave differently during gameplay.		
F3.1	Some enemies shall move faster than others.	Start a level.	Multiple enemies will spawn, some of them will be faster than the others
F3.2	Some enemies shall deal higher damage or have increased health compared to basic enemies.	The player gets hit by an enemy.	Different damage done to the player depending on the enemy.
F4.2	The game shall increase difficulty as the player progresses through rooms	Environment and enemies' power increases (e.g. more	The game difficulty increases.

		damage, different skills).	
F9	The enemies shall walk towards the player, but it should not be impossible for the player to run from them.	Enemy AI updates during gameplay. Player moves.	Enemies move in the direction of the player's current position. The player can move away from the enemies.
F6	The game shall allow the player to pick up items during gameplay.	The player interacts with an item found during the level.	The item disappears from the map and is added to the player's inventory.
F6.1	Collected items shall affect the player's gameplay, such as survivability or improving abilities.	The player collects an item that provides a gameplay effect like health boost or ability improvement.	The player's stats or abilities change accordingly(e.g. health increases or an ability becomes stronger).
Audio			
F5	The game shall provide audio feedback for important gameplay events.		
F5.1	Sound shall play when the player uses a special ability or when the player attacks and defeats an enemy.	The player performs the action.	A sound effect is played (respective to the action).
F5.2	Sound shall play when the player takes damage or is defeated by an enemy.	The player gets attacked.	A sound effect is played (respective to the action).
Misc.			
F2	The game shall allow the user to control the character by input from the keyboard and mouse.		
F2.1	Player movement shall be controlled by keyboard input.	Keyboard and mouse input.	The player's character responds correctly to the keyboard key binds mapping.
F2.2	Player direction (aim) shall follow the mouse pointer.	Mouse pointer movement.	Player front-view moves by tracking mouse pointer correctly.
F2.3	Main player attack action shall be performed by mouse left click.	Mouse left click input.	Player performs the attack (if successful, the enemy's health decreases).
F2.4	Player special attack shall be triggered by keys 'Q' and 'E'.	Input of keys 'E' or 'Q' via keyboard.	Activates special attack/ability.
F7.2	The game shall allow the player to be able to restart the level after death.	The player selects the restart option from the defeat screen.	The level restarts from the beginning with the player's health reset to its default value as well as enemies, items and timers return to their initial state.

F8	The game should spawn enemies at random locations on the map but not directly on top of the player.	Start a level.	Enemies appear at different valid spawn points within the room (e.g. not at the same position as the player or other objects).
F10	Enemies shall disappear after they die.	Kill an enemy	Enemy's corpse disappears
F14	The game shall spawn unbreakable static objects on the map but not directly on top of the player.	Start a level.	The objects are spawned at different locations on the map, excluding the one of the player.
F14.1	The objects shall block player or enemy movement.	The player or the enemies move.	The player or the enemy cannot go through the object.
F14.2	The objects shall remain in the same position for the entire duration of each level.	The object is spawned.	The object does not change its location after it is spawned.
F14.3	The objects shall block attacks from both the player and the enemies.	The player or the enemies attack.	The attack is blocked by the static object.
F14.4	The objects shall not be destroyed by any form of attack.	The object is attacked.	The object is not destroyed.
F16	The player shall be able to pause the game at any time, except during a cutscene.	The player selects to pause the game.	The pause menu shows up.
F16.1	The pause menu shall show options for continuing or restarting the game.	The player selects to continue or restart the game.	The game continues or is restarted.

4. Implementation Plans

4.1. Task List

Programming				
ID	Milestone	Dependencies	Deliverables	Deadline (End of Week)
P1	Program a core GUI.	None	Python files providing core functionality of GUI.	Week 16
P2	Program the core functionality of the rooms.	P1	Python files providing the core functionality of the rooms.	Week 17
P3	Program the core functionalities of the player.	P2, C3, G3	Python files providing the core functionality of the player.	Week 17
P4	Program the core functionalities of the enemies.	P2, C4, G4	Python files providing the core functionality of the enemies.	Week 18

P5	Program the skills.	P3	Python files providing the core functionality of the skills.	Week 18
P6	Program the weapons.	P3	Python files providing the core functionality of the weapons.	Week 19
P7	Program the items/upgrades.	P3, C5	Python files programing the core functionality items the items upgrades.	Week 19
P8	Upgrade the core GUI.	P1, C1, C2, G1	Python files upgrading the core functionality of the GUI.	Week 20
P9	Program timer.	P10	Python files providing the core functionality of timer.	Week 20
P10	Program waves of enemies.	P2, P4, G4	Python files programing the core functionality of the waves of enemies.	Week 21
P11	Program the boss.	P2	Python files programing the core functionality of the boss.	Week 21
P12	Upgrade skills design.	P5	Python files upgrading the core functionality of skills design.	Week 22
P13	Upgrade weapons design.	P6	Python files upgrading the core functionality of items design.	Week 22
P14	Program map obstacles.	P1	Python files providing the core functionality of map obstacles.	Week 22
P15	Program game flow.	P3, P10, P11	Python files providing the core functionality of game flow.	Week 22
P16	Program sound effect functionality.	P5, P6, P10, S1, S2, S3, S4,	Python files programing the core functionality the sound effect functionality.	Week 22
Content Design				
C1	Design 4 levels.	None	Creating the design for the levels that will be included in the game.	Week 17
C2	Design the character.	None	Creating the design for the main character.	Week 17
C3	Design different categories of enemies.	None	Creating the design for different enemies.	Week 18
C4	Design different items.	None	Creating the design for all items in the game.	Week 18
C5	Write the script for the story cutscenes.	None	Script for the story cutscenes	Week 19
Graphic Design				
G1	Produce/Find assets for the core GUI.	None	.PNG files for each required asset for the core GUI.	Week 18
G2	Produce/Find assets for the levels.	C1	.PNG files for each required asset for all levels.	Week 19
G3	Produce/Find assets for the main character.	C2	.PNG files for each required asset for the main character, including animations.	Week 16
G4	Produce/Find assets for each enemy.	C3	.PNG files for each required asset for all enemies, including animations.	Week 17
G5	Produce/Find assets for each item.	C4	.PNG files for each required asset for all items.	Week 20
Sound Design				

S1	Source a soundtrack for each level.	None	.WAV files for each required soundtrack.	Week 20
S2	Source sound effects for the main character animations.	None	.WAV files for each required sound effect.	Week 21
S3	Source sound effects for each enemy.	None	.WAV files for each required sound effect.	Week 21
S4	Source sound effects for the story cutscenes.	None	.WAV files for each required sound effect.	Week 22

All the members of the group will be equally working together to build the game. Some of the members are going to be more focused on the assets and 'creative' aspects of the project, such as the design of the characters, map, and other visuals, while other member will be more focused on the mechanics of the game and logic, such as the mechanics of the player, interaction with the items, and logic of the enemies.

Sava will be working on the logic of the enemies.

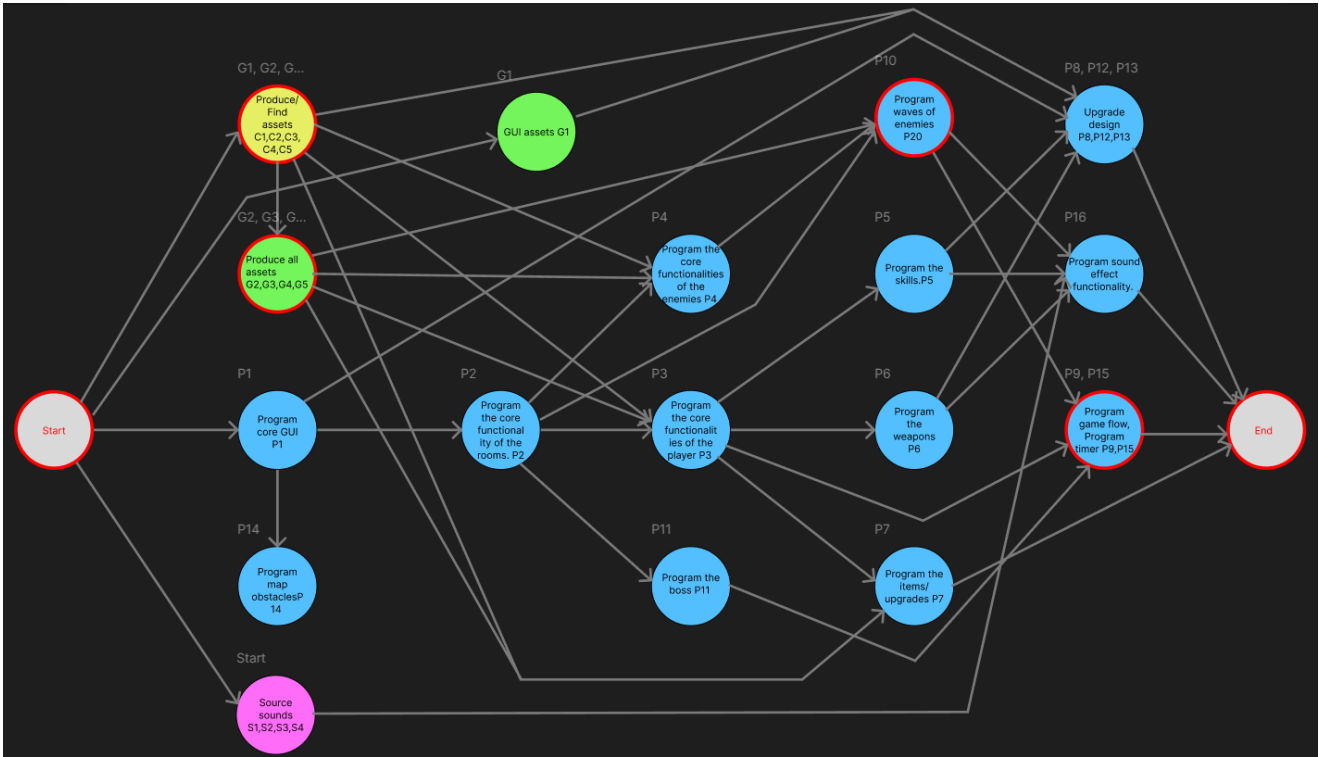
Joe will be working on the logic of the enemies and behavior of the map.

Sabrina will be working on the logic and movement of the player, and the design of the assets.

Ignacio will be working on the skills, items, and the design of the characters, maps, and music/sound effects.

Marcos will be working on the logic of enemies, general game flow, and environmental objects.

4.2. Activity Network with Critical Path



4.3. Gantt Chart

